

Heat circulation modelling : Oven case study

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M1AM

Variational Methods applied to modelling - Frédérique Charles and Clément Jourdana

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1. Direct problem

a) Variational formulation

Let Γ_t be the top wall, Γ_b the bottom door, and Γ_l, Γ_r the lateral walls. We fix the source intensities $\alpha = (\alpha_1, \dots, \alpha_6)$ and set $f = \sum_{i=1}^6 \alpha_i 1_{C_i}$. Since the C_i have finite measure, $f \in L^2(\Omega)$. The strong problem is

$$\begin{cases} -\operatorname{div}(\kappa \nabla T) = f & \text{in } \Omega \\ T = T_c & \text{on } \Gamma_t \\ \partial_n T = 0 & \text{on } \Gamma_l \cup \Gamma_r \\ \partial_n T + h(T - T_e) = 0 & \text{on } \Gamma_b \end{cases} \quad (1)$$

Following the course treatment of mixed boundary value problems, the Dirichlet condition is put in the space. Let $V = \{v \in H^1(\Omega) \mid \gamma_0 v = 0 \text{ on } \Gamma_t\}$ and $K = \{w \in H^1(\Omega) \mid \gamma_0 w = T_c \text{ on } \Gamma_t\}$.

Assume first that T is regular. For $v \in V$, multiply the equation by v and integrate over Ω :

$$\int_{\Omega} f v \, dx = \int_{\Omega} -\operatorname{div}(\kappa \nabla T) v \, dx. \quad (2)$$

Green's formula gives

$$\int_{\Omega} -\operatorname{div}(\kappa \nabla T) v \, dx = \int_{\Omega} \kappa \nabla T \cdot \nabla v \, dx - \int_{\partial\Omega} \kappa \partial_n T v \, ds. \quad (3)$$

We now split the boundary term. On Γ_t , $v = 0$ by definition of V . On $\Gamma_l \cup \Gamma_r$, the walls are insulated, so $\partial_n T = 0$. On Γ_b , the Robin condition gives $\partial_n T = -h(T - T_e)$, hence $-\kappa \partial_n T v = \kappa h T v - \kappa h T_e v$. Therefore

$$\int_{\Omega} \kappa \nabla T \cdot \nabla v \, dx + \int_{\Gamma_b} \kappa h T v \, ds = \int_{\Omega} f v \, dx + \int_{\Gamma_b} \kappa h T_e v \, ds. \quad (4)$$

This identity only involves first derivatives of T and is the weak formulation. For $u, v \in H^1(\Omega)$, define

$$a(u, v) = \int_{\Omega} \kappa \nabla u \cdot \nabla v \, dx + \int_{\Gamma_b} \kappa h u v \, ds. \quad (5)$$

The weak problem is thus: find $T \in K$ such that, for all $v \in V$,

$$a(T, v) = \int_{\Omega} f v \, dx + \int_{\Gamma_b} \kappa h T_e v \, ds. \quad (6)$$

We cannot directly apply Lax-Milgram on K because K is affine, not vectorial. Choose any lifting $G \in H^1(\Omega)$ such that $\gamma_0 G = T_c$ on Γ_t ; for instance the constant function $G = T_c$. We write $T = U + G$, where $U \in V$. Then T solves the weak problem if and only if $U \in V$ satisfies $a(U, v) = \ell(v)$ for all $v \in V$, with

$$\ell(v) = \int_{\Omega} f v \, dx + \int_{\Gamma_b} \kappa h T_e v \, ds - a(G, v). \quad (7)$$

The term $a(G, v)$ is simply the contribution of the lifting of the non-homogeneous Dirichlet condition. It appears when replacing T by $U + G$ in $a(T, v)$.

The space V is a closed subspace of $H^1(\Omega)$, hence a Hilbert space for the H^1 norm. We now check the three assumptions of the Lax-Milgram theorem.

First, a is continuous on $V \times V$. Indeed, $\kappa \in L^\infty(\Omega)$ and the trace theorem gives $\|\gamma_0 v\|_{L^2(\Gamma_b)} \leq C\|v\|_{H^1(\Omega)}$. Thus there is $M > 0$ such that $|a(u, v)| \leq M\|u\|_{H^1(\Omega)}\|v\|_{H^1(\Omega)}$.

Second, a is coercive on V . Since $\kappa \geq 1$, $a(v, v) \geq \|\nabla v\|_{L^2(\Omega)}^2$. By the Poincaré inequality on the space of functions with zero trace on Γ_t , there exists $C_P > 0$ such that $\|v\|_{H^1(\Omega)} \leq C_P\|\nabla v\|_{L^2(\Omega)}$. Hence $a(v, v) \geq C_P^{-2}\|v\|_{H^1(\Omega)}^2$.

Third, ℓ is continuous on V . The volume term is controlled by Cauchy-Schwarz, the boundary term by the trace theorem, and the lifting term by the continuity of a .

By Lax-Milgram, there is a unique $U \in V$, hence a unique weak solution $T = U + G \in K$.

b) P1 finite element approximation

Let $\mathcal{T}_h$ be a triangulation of Ω . The P_1 finite element space is

$$V_h = \{v_h \in C^0(\overline{\Omega}) \mid v_h|_K \text{ is affine for all } K \in \mathcal{T}_h, v_h = 0 \text{ on } \Gamma_t\}. \quad (8)$$

With $G_h = T_c$, we look for $T_h = U_h + G_h$ with $U_h \in V_h$ such that $a(U_h, v_h) = \ell(v_h)$ for all $v_h \in V_h$.

Let $(\varphi_1, \dots, \varphi_N)$ be the usual nodal basis of V_h . Writing $U_h = \sum_{j=1}^N U_j \varphi_j$ and testing with each φ_i gives the linear system $A_h U = F_h$, with $(A_h)_{ij} = a(\varphi_j, \varphi_i)$ and $(F_h)_i = \ell(\varphi_i)$. The matrix is sparse because each hat function has local support. By applying Lax-Milgram on the finite-dimensional Hilbert space V_h , this discrete problem has a unique solution.

c) Stationary simulation

The script **direct.edp** implements the P_1 method for the stationary direct problem. For the test run, we used the prescribed six source locations and the known intensities $\alpha_i = 250000$ for all i . The direct simulation gives $\min T_h = 343K$, $\max T_h = 507.137K$, and $\overline{T}_{h_S} = 422.015K$.

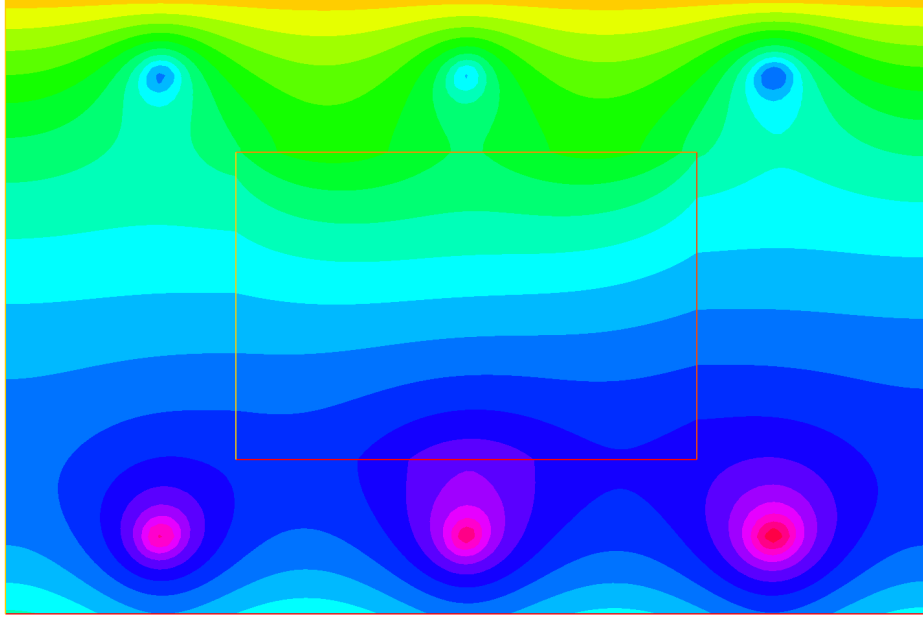


Figure 1 - Stationary temperature distribution for the direct problem, with the cooking region S outlined. Warmer colors indicate higher temperature.

The figure shows hot spots around the resistors and a non-uniform temperature in S , which motivates the optimization problem of Part 2.

d) Warm-up phase

To describe the warm-up, we add the time derivative and consider $\partial_t T - \operatorname{div}(\kappa \nabla T) = f$ in $\Omega \times]0, T_f[$, with the same boundary conditions as before. We use the following implicit Euler scheme: given T_h^n , find $T_h^{n+1} \in G_h + V_h$ such that, for all $v_h \in V_h$,

$$\int_{\Omega} \frac{T_h^{n+1} v_h}{\delta} dx + a(T_h^{n+1}, v_h) = \int_{\Omega} \frac{T_h^n v_h}{\delta} dx + \int_{\Omega} f v_h dx + \int_{\Gamma_b} \kappa h T_e v_h ds. \quad (9)$$

The script **warmup.edp** implements this scheme with $T_h^0 = T_e$, over 200 time steps on the interval $[0, 1.25]$. With the same source intensities as in **direct.edp**, the mean temperature in S goes from $295.491K$ at $t = 0.00625$ to $421.986K$ at $t = 1$, and reaches $422.012K$ at $t = 1.25$. This is consistent with the convergence of the parabolic problem toward the stationary direct solution.

2. Optimization problem

a) Linear decomposition of the solution

For a coefficient vector $\alpha = (\alpha_1, \dots, \alpha_6)$, set $f_\alpha = \sum_{i=1}^6 \alpha_i 1_{C_i}$. The direct strong problem is obtained from Part 1 by replacing f with f_α .

Define T_0 as the solution with no internal source,

$$\begin{cases} -\operatorname{div}(\kappa \nabla T_0) = 0 & \text{in } \Omega \\ T_0 = T_c & \text{on } \Gamma_t \\ \partial_n T_0 = 0 & \text{on } \Gamma_l \cup \Gamma_r \\ \partial_n T_0 + h(T_0 - T_e) = 0 & \text{on } \Gamma_b \end{cases} \quad (10)$$

and, for $i = 1, \dots, 6$, define T_i as the unit response of source C_i ,

$$\begin{cases} -\operatorname{div}(\kappa \nabla T_i) = 1_{C_i} & \text{in } \Omega \\ T_i = 0 & \text{on } \Gamma_t \\ \partial_n T_i = 0 & \text{on } \Gamma_l \cup \Gamma_r \\ \partial_n T_i + h T_i = 0 & \text{on } \Gamma_b. \end{cases} \quad (11)$$

T_0 is the temperature when all resistors are switched off, and T_i is the temperature response generated by source i alone with homogeneous boundary data. Let

$$T^*(\alpha) = T_0 + \sum_{i=1}^6 \alpha_i T_i. \quad (12)$$

By linearity of $-\operatorname{div}(\kappa \nabla \cdot)$ and of the boundary operators, $T^*(\alpha)$ satisfies the same strong problem as $T(\alpha)$. By uniqueness of the solution, we conclude that

$$T(\alpha, x) = T_0(x) + \sum_{i=1}^6 \alpha_i T_i(x). \quad (13)$$

b) Linear formulation of the optimization problem

Set $e_0(x) = T_0(x) - T_s$. We have

$$J(\alpha) = \frac{1}{2} \int_S |T(\alpha, x) - T_s|^2 dx = \frac{1}{2} \int_S \left| e_0 + \sum_{j=1}^6 \alpha_j T_j \right|^2 dx. \quad (14)$$

For each $i = 1, \dots, 6$, differentiating under the integral gives

$$\partial_{\alpha_i} J(\alpha) = \int_S \left(e_0 + \sum_{j=1}^6 \alpha_j T_j \right) T_i dx. \quad (15)$$

Setting the gradient to 0 yields

$$\sum_{j=1}^6 \left(\int_S T_i T_j dx \right) \alpha_j^* = \int_S (T_s - T_0) T_i dx, \quad i = 1, \dots, 6. \quad (16)$$

Therefore the optimization problem is reduced to the linear system $A\alpha^* = b$, where

$$A_{ij} = \int_S T_i T_j \, dx, \quad b_i = \int_S (T_s - T_0) T_i \, dx. \quad (17)$$

The matrix A is symmetric and positive semidefinite because, for every $\beta \in \mathbb{R}^6$, $\beta^T A \beta = \int_S \left| \sum_{i=1}^6 \beta_i T_i \right|^2 dx \geq 0$. To conclude that A is positive definite, we need the linear independence of the restrictions $T_1|_S, \dots, T_6|_S$ in $L^2(S)$. In the present geometry this can be proved by using the transmission conditions and unique continuation for harmonic functions. Since this argument is outside the main scope of the course, we give it in the appendix. It shows that A is definite thus invertible. We can conclude that the linear system, thus the optimization problem, has a unique solution.

c) Simulation results

The script **optimization.edp** implements the previous construction with P_1 finite elements. It computes T_0 , then the six functions T_i , assembles the discrete normal system $A^h \alpha^h = b^h$, solves it, and finally performs a direct simulation with the resulting source term.

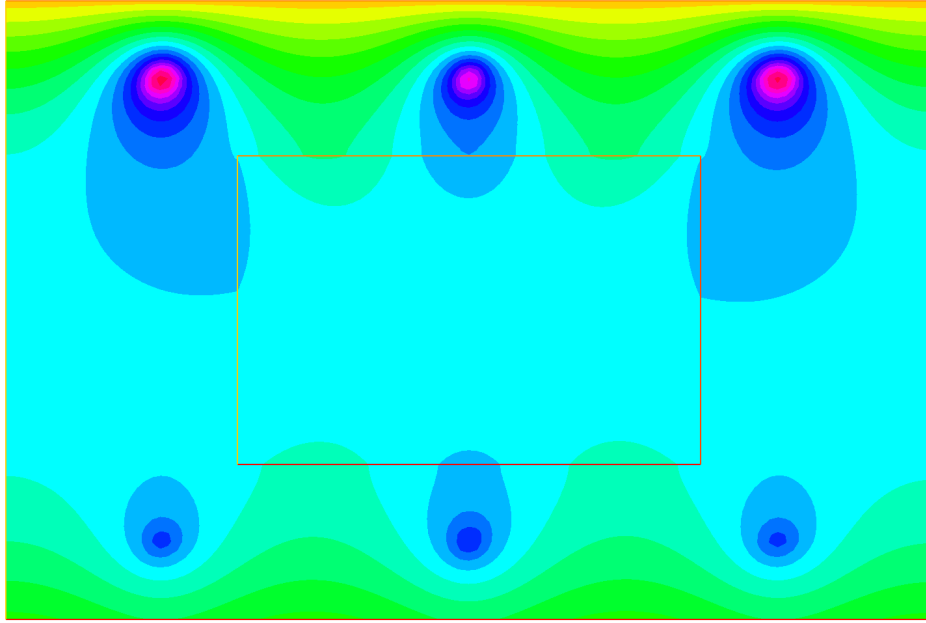


Figure 2 - Optimized temperature distribution.

For $T_s = 400K$, the direct simulations with the optimized coefficients give

$$\bar{T}_S = 399.9K \quad \text{and} \quad \sqrt{\frac{1}{|S|} \int_S |T - \bar{T}_S|^2 dx} = 2.49K. \quad (18)$$

The mean temperature in S is therefore very close to the target $400K$, and the L^2 error in S is about $2.49K$.

The result is symmetric under the reflection $x \rightarrow -x$, as the domain, the coefficient κ , the region S , the source layout, and the boundary conditions are invariant under this reflection. There is no analogous symmetry in the vertical direction: the top wall is fixed at $T_c = 343K$, whereas the

bottom wall exchanges heat with an exterior temperature $T_e = 293K$, so the bottom sources remain cooler than the top ones.

d) Source optimization

We then tested different source layouts while keeping the same radius $r_C = 0.05$ and the same target temperature $T_s = 400K$. The best tested configuration uses eight sources: five on the upper row and three on the lower row,

$$(-1, 0.75), (-0.5, 0.75), (0, 0.75), (0.5, 0.75), (1, 0.75) \quad (19)$$

and

$$(-1, -0.75), (0, -0.75), (1, -0.75). \quad (20)$$

We chose this distribution because the original optimization already required larger coefficients on the upper sources. Adding more sources on the upper side therefore gives finer control where it is most useful. We also keep symmetry with respect to $x = 0$, since the oven geometry, the cooking region, and the target temperature are symmetric in the horizontal direction.

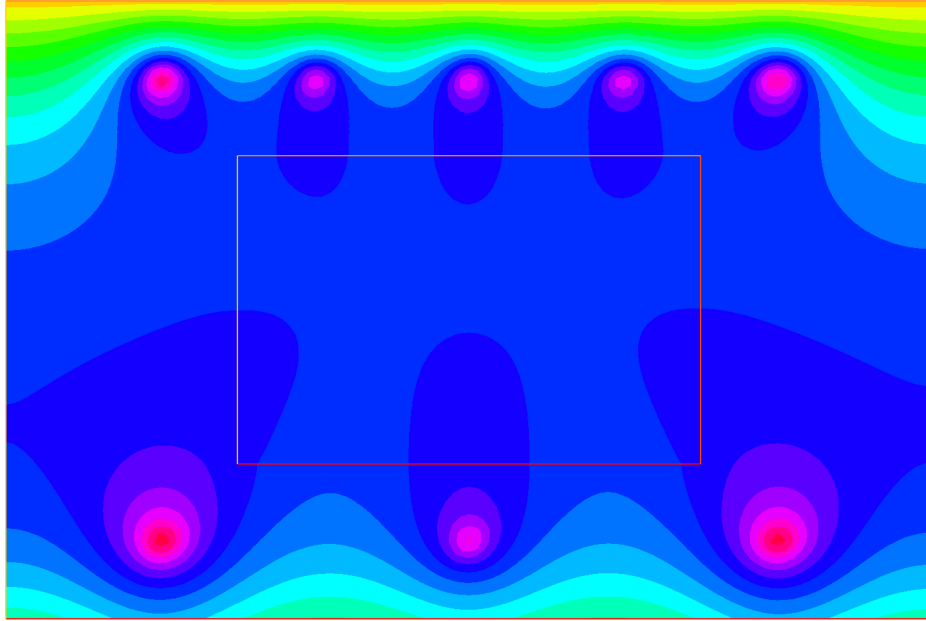


Figure 3 - Optimized temperature distribution for the proposed eight-source layout.

The script `optimized_sources.edp` gives

$$\bar{T}_S = 399.994K \quad \text{and} \quad \sqrt{\frac{1}{|S|} \int_S |T - T_s|^2 dx} = 0.804K. \quad (21)$$

For comparison, the initial six-source layout gave an error approximately equal to $2.49K$. The proposed layout therefore reduces the error significantly while keeping all optimized source intensities positive thus physically feasible. We also observe the expected smoother distribution of intensities across sources.

3. | Inverse problem

a) Detection method

We assume that the normal operating state of the oven is known. Thus, from a measured temperature T_m , we first subtract the no-anomaly background T_0 . The anomaly $U = T_m - T_0$ satisfies the same homogeneous boundary conditions as the responses T_i from the optimization part, but with one unknown source $\alpha 1_{C(z)}$, where z is the unknown center.

For a candidate center z , let u_z be the solution with unit intensity:

$$a(u_z, v) = \int_{C(z)} v \, dx \quad \text{for all } v \in V. \quad (22)$$

If B_M denotes the measurement operator, the predicted measurement is $\alpha B_M u_z$. We use two choices: $B_M w = w$ on $\Gamma_l \cup \Gamma_r$, which corresponds to measuring the temperature on the vertical walls, and $B_M w = \partial_n w$ on Γ_t , which corresponds to measuring the normal derivative on the top wall.

For each candidate z , the best intensity is obtained explicitly by a one dimensional least-squares fit:

$$\alpha(z) = \frac{(d_M, B_M u_z)_M}{(B_M u_z, B_M u_z)_M}, \quad (23)$$

where $d_M = B_M(T_m - T_0)$ and $(\cdot, \cdot)_M$ is the L^2 product on the measured boundary. The remaining relative residual is

$$r(z) = \sqrt{\frac{(d_M - \alpha(z) B_M u_z, d_M - \alpha(z) B_M u_z)_M}{(d_M, d_M)_M}}. \quad (24)$$

The numerical inverse problem is then reduced to a finite search over candidate centers: choose the z minimizing $r(z)$, and take $\alpha(z)$ as the recovered intensity.

b) Numerical results

The script **inverse.edp** implements this dictionary search. We generated a synthetic anomaly with center $z^* = (0.4, -0.3)$, radius 0.05, and intensity $\alpha^* = 5 \times 10^5$. The candidate grid is 29×19 points in $[-1.4, 1.4] \times [-0.9, 0.9]$; this grid contains the true center.

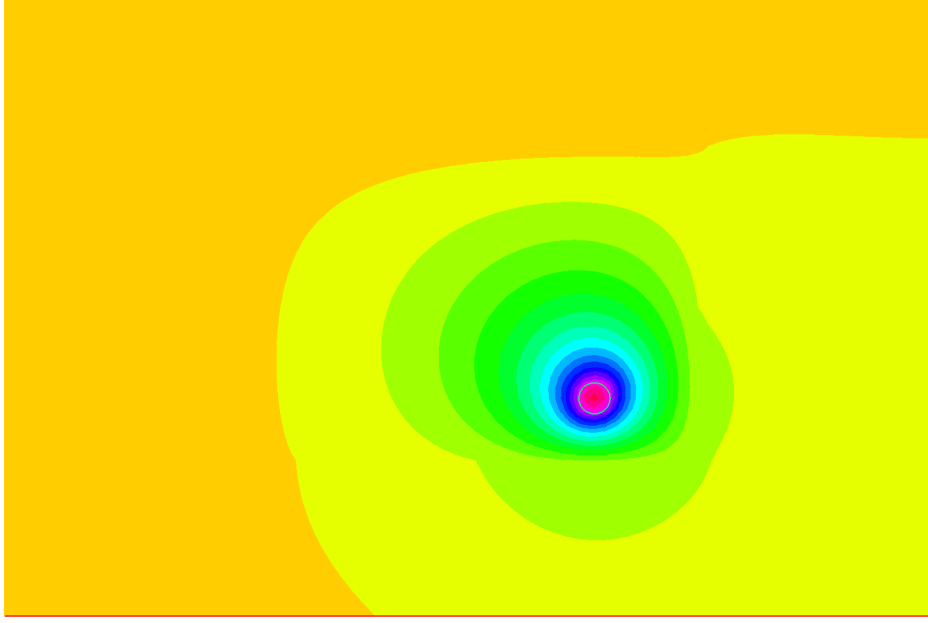


Figure 4 - Temperature distribution with unknown source and recovered center drawn in blue.

Using the temperature on the left and right walls, the algorithm returns $z = (0.4, -0.3)$, $\alpha = 5 \times 10^5$, with relative residual 0. Using $\partial_n T$ on the top wall, it also returns $z = (0.4, -0.3)$ and $\alpha = 5 \times 10^5$, with relative residual 1.39×10^{-8} . The small non-zero residual in the second case comes from the finite element evaluation of the boundary normal derivative.

These results confirm that, in the noiseless synthetic setting, both proposed measurement types are sufficient to detect the anomaly when the candidate grid contains the true source. In practice, the inverse problem remains sensitive to noise and to the grid resolution because the heat equation smooths the source before it is observed at the boundary. If the true source is not on the grid, the same method identifies the nearest candidate response; refining the grid then improves the location estimate.

Appendix

We prove¹ that $T_1|_S, \dots, T_6|_S$ are linearly independent in $L^2(S)$ for the project geometry. The source disks are outside S , their closures are pairwise disjoint, and none of them touches ∂S . The coefficient κ is constant equal to 1 in S and constant equal to 10 outside S .

Assume that $\sum_{i=1}^6 \beta_i T_i = 0$ in S , and set $W = \sum_{i=1}^6 \beta_i T_i$. Then $-\operatorname{div}(\kappa \nabla W) = \sum_{i=1}^6 \beta_i 1_{C_i}$ in Ω , with homogeneous boundary conditions. Since the sources are away from ∂S , W is harmonic on both sides of each open side segment of ∂S .

On the side of S , $W = 0$, hence both its trace and normal derivative vanish there. The standard transmission conditions for a weak solution of $-\operatorname{div}(\kappa \nabla W) = F$ give continuity of W and of the normal flux $\kappa \partial_n W$ across ∂S . Thus, on the exterior side of any open side segment of ∂S , we also have $W = 0$ and $\partial_n W = 0$.

Near such a segment, extend the exterior harmonic function by zero across the segment. The zero Cauchy data make this extension weakly harmonic; by elliptic regularity it is harmonic. It vanishes on a non-empty open set, so by analyticity of harmonic functions it vanishes in a whole neighbourhood outside S . The exterior region obtained by removing $\overline{S}$ and the closed disks $\overline{C_i}$ is connected, and W is harmonic there. Analytic continuation therefore gives $W = 0$ throughout this exterior region.

Now fix one disk C_i . Since $W = 0$ outside the disk, the trace of W and the exterior normal derivative vanish on ∂C_i . There is no boundary measure in the equation, so the interior normal derivative also vanishes. Inside C_i , we have $-10\Delta W = \beta_i$. Integrating over C_i gives

$$\beta_i |C_i| = -10 \int_{C_i} \Delta W \, dx = -10 \int_{\partial C_i} \partial_n W \, ds = 0. \quad (25)$$

Since $|C_i| > 0$, we get $\beta_i = 0$. Thus the restrictions $T_i|_S$ are linearly independent in $L^2(S)$.

Note that the circular shape is not essential. This proofs works for any finite family of pairwise disjoint Lipschitz open source regions of positive measure, compactly contained in $\Omega - \overline{S}$, provided $\Omega - (\overline{S} \cup \cup_i \overline{C_i})$ is connected.

¹Please note that this proof was not AI-generated. One of the authors has studied complex and harmonic analysis previously and was more than happy to apply it to this context :)